

A Coefficient- and Phase-Resolved Bridge across Noncommuting Limits of Lamb-Wave ZGV Splitting

Abstract

Breaking continuous rotational symmetry changes the dimension of a stationary set and can make the symmetry-restoring and long-time limits noncommuting. In isotropic Lamb waves, a finite-wavenumber zero-group-velocity (ZGV) critical ring produces a radial $t^{-1/2}$ response, whereas cubic anisotropy resolves the ring into isolated Morse points with an ultimate t^{-1} response. These endpoints were known. Our dated literature audit found no prior plate calculation joining their full-elastic coefficients, normalization, and signature phases under the controls used here. We derive that bridge for a selected simple, uniformly gapped branch. Generalized-eigenvalue sensitivities determine the angular and radial coefficients, and a Morse–Bott reduction proves 4 minima and 4 saddles for sufficiently small nonzero perturbations. A finite-resolution calculation yields eight resolved, refinement-stable roots without asserting continuum exhaustion. After removal of the shifted carrier $\Omega_0 + \varepsilon V_0$, direct branch quadrature follows $t^{-1/2} J_0(\varepsilon V_4 t)$; a separately evaluated fixed-anisotropy slice approaches the exact-Hessian Morse sum, and a proved growing-parameter overlap matches their coefficients and phases. The comparisons are performed without fitted alignment. Their largest accumulated inter-resolution phase-discrepancy estimator is 1.28586×10^{-6} rad; this empirical quantity is not a continuum error bound. The computation-only result is restricted to a linear, lossless infinite plate, the declared local annulus, and the specified source and weak-anisotropy classes.

1 Introduction

Finite-wavenumber zero-group-velocity (ZGV) Lamb states are central to long-lived localization in plates. They retain a nonzero in-plane wavelength while a stationary guided-wave dispersion suppresses the leading transport of energy away from the source, producing local vibration and a temporal decay law controlled by the geometry of the stationary set. Laser-ultrasonic excitation and local ZGV responses are established phenomena [17, 18]. In an isotropic plate, rotational symmetry turns a radial finite-wavenumber stationary point into a continuous ZGV critical ring, and radial stationary phase gives the established $t^{-1/2}$ decay for a nonnodal local response [15, 19]. Neither the ZGV critical ring nor this exponent is at issue here as a new effect; they provide the known isotropic endpoint whose loss of directional degeneracy we examine.

Elastic anisotropy provides the other established endpoint. Anisotropic-plate studies established directional ZGV shifts [20]. Calculations also found multiple stationary points [8, 10], and isolated anisotropic ZGV points were previously reported. More specifically, four minima and four saddles, together with their interference, were previously reported for a cubic silicon plate [11]; subsequent work described their directional stationary-phase organization and unfolding [13]. Algorithms for computing isolated ZGV points in the full guided-wave spectrum are likewise established [12, 16]. The present calculation therefore does not claim the isolated points, their alternating four-plus-four organization, or the associated interference as new observations. Its question is how these known anisotropic objects emerge in a controlled limit from the known isotropic ZGV critical ring.

A plate-specific question remains open in the literature examined here: how to construct a coefficient-resolved connection between that geometry and the temporal asymptotics of the full Lamb eigenproblem. Morse–Bott critical manifolds and their perturbations are established mathematics [2, 3], while two-dimensional stationary phase has long been applied to isolated guided-wave points [6, 9, 21]. Broken-symmetry uniform approximations in other wave settings also connect continuous families to isolated objects through a Bessel modulation [4, 5, 7]. The required connection must retain full-elastic angular and radial coefficients, a derivative-level remainder, source normalization, and a phase-controlled temporal comparison. This missing link, rather than a new general Morse or Bessel theorem, is the focus of the analysis.

Here we connect the known ZGV critical ring to the anisotropic endpoint. The reduction treats weak cubic anisotropy on a simple tracked Lamb branch. Full-elastic generalized-eigenvalue sensitivities determine the angular coefficient V_4 and the radial coefficient B without fitting. Critical-point geometry, Cartesian Hessian inertia, and annular index checks then test the alternating local Morse set and its perturbative scaling inside a declared local annulus. Uniform Bessel asymptotics connect the ring response to the isolated-point regime, while the fixed-anisotropy endpoint retains exact critical frequencies and Hessians. Finally, validated spectral computations and an accumulated inter-resolution phase-discrepancy estimator test the entire chain without fitted amplitude, phase, frequency, or time shifts. The study is computation-only: it concerns one selected simple branch and a controlled weak-anisotropy family; the finite-anisotropy silicon stress test is separate from the asymptotic evidence.

2 The Isotropic ZGV Ring

2.1 Exact Rayleigh–Lamb reference

We restrict the isotropic reference calculation to the selected symmetric S_1 branch and the configured local window around its zero-group-velocity (ZGV) state. Within that scope, the traction-free Rayleigh–Lamb determinant is desingularized before root finding, so a vanishing shear thickness wavenumber cannot be mistaken for a physical root. With $\kappa = kh$ and $\Omega = \omega h/c_T$, simultaneously solving $D_S = D_{S,k} = 0$ gives the dimensionless location $\kappa_0 = 0.804217$ and frequency $\Omega_0 = 2.85176$; implicit differentiation gives the positive dimensionless radial curvature $a = d^2\Omega/d\kappa^2 = 1.19686$. These quantities characterize one simple local branch, not all symmetric, antisymmetric, or shear-horizontal branches of the plate.

The reference starts from the physical face-traction matrix rather than a quotient form of the classical dispersion relation [14]. Rescaling the shear potential produces the regular determinant in Eq. (22), which remains finite through the shear cutoff. The nonzero frequency derivative then identifies a smooth scalar branch, while Eqs. (30) and (33) convert the double condition on D_S into zero radial group velocity and a directly differentiated curvature. The independently refined values declared in Eq. (35) therefore come from the exact isotropic determinant and are not fitted to the subsequent spectral calculation.

Near the critical radius, the branch is described by $\Omega = \Omega_0 + a(\kappa - \kappa_0)^2/2$ to leading order. The positive sign of a makes the selected state a radial minimum, consistent with the familiar local ZGV geometry and its established role in long-lived plate responses [19]. This local quadratic statement is used only inside the configured window; it does not replace the exact branch away from the reference point.

2.2 Morse–Bott geometry

Within a sufficiently thin local annulus on the selected isotropic branch, rotational invariance lifts the radial ZGV state to the critical ring $\Gamma_0 = \{\boldsymbol{\kappa} \in \mathbb{R}^2 : |\boldsymbol{\kappa}| = \kappa_0\}$, where $\boldsymbol{\kappa} = h\mathbf{k}^{\text{phys}}$. We use the Morse–Bott classification to state its degeneracy precisely: the radial normal direction is nondegenerate, whereas the tangential zero direction is enforced by continuous rotational symmetry. This is a plate-specific classification of the familiar isotropic ZGV circle, not an assertion that the circle itself is newly found, and it requires the local branch to remain simple and uniformly separated from neighbouring modes.

The Cartesian Hessian with respect to $\boldsymbol{\kappa}$ on the ring is $a\mathbf{e}_r\mathbf{e}_r^\top$, with spectrum $\{a, 0\}$, as shown in Eq. (42). Its kernel is therefore exactly the tangent space of Γ_0 , while its restriction to the radial normal space is the nonsingular scalar form $[a]$ in Eq. (44). These two facts meet the normal-nondegeneracy condition of Theorem 1.1 [3]. They also distinguish one connected critical manifold from a collection of unrelated isolated points. Cases with zero radial curvature or a closing eigengap lie outside the theorem.

Figure 1 summarizes the resulting geometry and previews its weak-cubic unfolding. The isotropic contours share a stationary circle, whereas the anisotropic panel places the refined full-wave minima and saddles inside the declared local annulus. Their count and classification are justified with the full anisotropic calculation below; only the final dimension-and-decay map is conceptual. Related anisotropic ZGV organization and temporal interference have already been reported [11]; here the diagram serves only as a roadmap to the coefficient-resolved calculation.

2.3 Independent spectral verification

We next test the exact reference with an independent thickness-resolved calculation, restricted to the same configured local window and tracked spectral branch. The generalized Hermitian eigenproblem uses traction-free natural boundary conditions and no Rayleigh–Lamb determinant. Thus its agreement with the exact root checks a distinct numerical representation of the elastic field. This is a computation-only verification rather than an experiment: no measured resonance, fitted material parameter, or laboratory observable enters the comparison.

The spectral weak form of Eq. (160) resolves all displacement components through the thickness. Branch continuity is enforced by the mass-weighted overlap procedure in Algorithm 167, and queries are rejected when the local separation condition in Eq. (168) is not met. Across the declared order sweep, the normalized eigen-residual of Eq. (164), the mass-orthogonality defect of Eq. (165), and the exact-reference errors remain small while the relative eigengap stays open. This simultaneously checks branch identity, field resolution, and the simple-eigenvalue hypothesis used by Theorem 1.1; it is not a claim about branches outside the tracked window. Spectral guided-wave discretization itself is established methodology [1].

Figure 2 separates the four pieces of this check. The tracked symmetric branch has a local minimum at the computed (0.804217, 2.85176) reference; the exact branch and its quadratic form with curvature 1.19686 agree over the declared local comparison interval. The convergence panel reports errors and eigengap together, preventing apparent frequency agreement from masking branch coalescence. Finally, the mass-normalized mode panel shows the through-thickness component magnitudes and their squared-displacement proxy; because it plots magnitudes, it is not used as a signed parity test. These mutually consistent diagnostics support the Morse–Bott statement only for the selected uniformly gapped local branch.

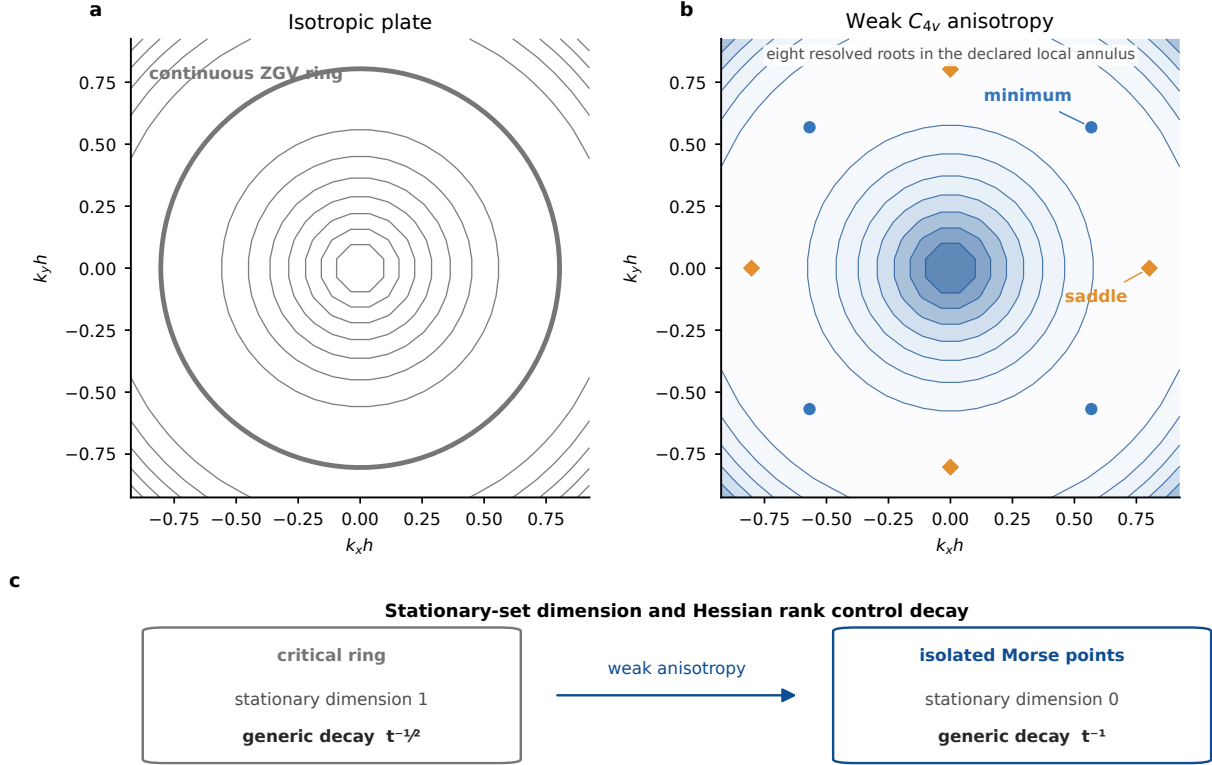


Figure 1: Mechanism overview on the selected branch. **a**, Thin grey contours show the isotropic full-wave local-frequency surface, and the solid blue curve marks the continuous ZGV ring at the exact isotropic radius κ_0 . **b**, Filled contours and blue contour lines show the weakly anisotropic full-wave surface at the declared positive perturbation; blue circles and orange diamonds mark the computed, independently refined Morse minima and saddles, respectively, within the declared local annulus. Marker shape duplicates colour, and no point is inferred from or displaced relative to the contour field. **c**, Conceptual summary of the change from a one-dimensional critical ring with generic $t^{-1/2}$ decay to isolated Morse points with generic t^{-1} decay; this panel is explanatory rather than numerical, with the computed asymptotic slices tested in Fig. 6. This mechanism overview does not carry the numerical realization claim; the finite-resolution full-wave search, Cartesian Hessians, and stability checks are reported in Fig. 4. Wavevector components are normalized as $k_x h$ and $k_y h$, frequencies as $\Omega = \omega h / c_T$, and panels **a** and **b** use common contour levels and axes. Source data: `data/source_data/figure_01`.

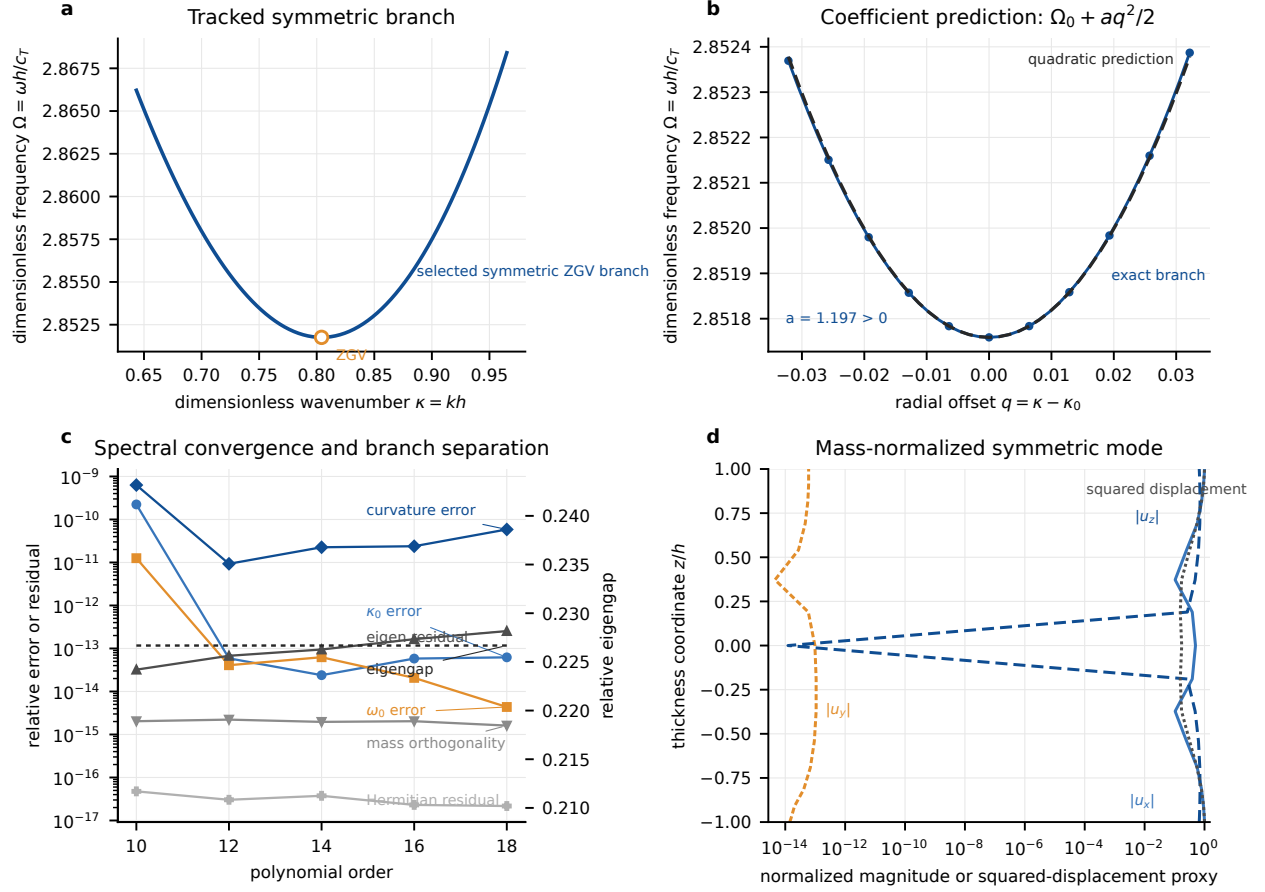


Figure 2: Independent numerical verification of the isotropic reference. **a**, The solid curve is the independently discretized full-wave symmetric branch; the open orange circular symbol marks the exact Rayleigh–Lamb reference (κ_0, Ω_0) . **b**, The solid curve with circular markers is the local full-wave branch, whereas the dashed line is the coefficient prediction $\Omega_0 + a(\kappa - \kappa_0)^2/2$. The positive curvature a is obtained from exact implicit differentiation, not from a fit to the plotted samples. **c**, Coloured marker–lines on the left axis report relative κ_0 , Ω_0 , and curvature errors together with the normalized eigen residual, Hermitian defect, and mass-orthogonality defect; the short-dashed line on the right axis is the relative eigengap. **d**, Through-thickness profiles of the mass-normalized mode: solid, short-dashed, and long-dashed curves denote $|u_x|$, $|u_y|$, and $|u_z|$, divided by their common global maximum. The neutral dotted curve is the squared-displacement proxy $\sum_i |u_i|^2$, normalized by its own maximum; it is neither a signed-parity nor a strain-energy diagnostic. Here $\kappa = kh$, $\Omega = \omega h/c_T$, and z/h is the normalized thickness coordinate. Source data: `data/source_data/figure_02`.

3 Symmetry-Breaking Normal Form

3.1 Full-elastic angular sensitivities

Within the declared one-parameter cubic family, the angular and radial coefficients were obtained from the full traction-free generalized eigenproblem, rather than fitted from the eventual critical-point locations. The general Hermitian-pencil derivative in Eq. (50) gave the ring sensitivity $V(\theta)$, while the complete radial derivative in Eq. (60) retained the differentiated mode, equivalently the reduced-resolvent contribution, in $B(\theta)$. This calculation is restricted to the simple tracked branch and the declared cubic perturbation; it does not claim a coefficient formula for an arbitrary anisotropic plate.

The resulting angular potential contained the derived pure fourfold term of Eq. (75), with signed coefficient $V_4 = 0.0140638$. Here V_4 is the coefficient of the perturbation parameter, whereas the physical fourfold frequency shift is εV_4 , as fixed by Eq. (76); the perturbation parameter is therefore counted once. Centered full-wave finite differences independently recovered both $V(\theta)$ and $B(\theta)$. Figure 3 shows the harmonic reconstruction, the coefficient-to-shift convention, and the finite-difference closure. These checks make the full-elastic coefficient closure, rather than symmetry alone, the quantitative input to the unfolding.

3.2 Normal form and splitting theorem

Under the controlled weak-anisotropy hypotheses, eliminating the radial stationarity equation reduced the local dispersion to an angular problem in the declared local annulus. Theorem 2.1 and Eq. (78) retain the radial quadratic, the angular sensitivity, the mixed radial–angular coefficient, and a derivative-level remainder. The implicit radial graph in Eq. (81) is unique in this neighborhood; substituting it gives the reduced angular frequency of Eq. (82). These steps require a simple separated eigenvalue and a nonzero radial curvature.

The cubic splitting theorem in Theorem 2.2 then placed the critical angles and radial corrections according to Eq. (101). For every sufficiently small nonzero ε with $V_4 \neq 0$, the theorem yields 4 Cartesian Morse minima and 4 Cartesian Morse saddles in alternating angular order. This exact local count depends on the derived leading harmonic and controlled remainder, not on C_{4v} symmetry by itself and not on a global survey of every dispersion branch.

The retained isotropic radial curvature is positive. Consequently, the sign of the tangential curvature distinguishes the Cartesian Hessian classes: two positive eigenvalues give a minimum, whereas one positive and one negative eigenvalue give a saddle. This inertia statement follows from the exact polar-to-Cartesian Hessian in Eq. (87) and its Schur complement in Eq. (89). A direction with negative angular curvature is thus not a maximum. Reversing the sign of εV_4 exchanges the minimum and saddle angular roles while preserving their alternating organization. The four-minimum/four-saddle pattern is known in anisotropic ZGV calculations [11, 13]; the contribution here is the full-elastic coefficient closure together with a controlled local explanation of why that pattern follows for this perturbation family.

3.3 Eight resolved critical points

We restricted the full-wave search to the declared local annulus on the tracked full-elastic branch. The thickness discretization retained all three displacement components. Within this scope, the search found eight resolved roots, comprising 4 minima and 4 saddles. Every accepted root satisfied the gradient-resolution condition associated with Eq. (171), and its Cartesian Hessian eigenvalues remained separated from their numerical discrepancy estimate. Candidate-grid refinement preserved

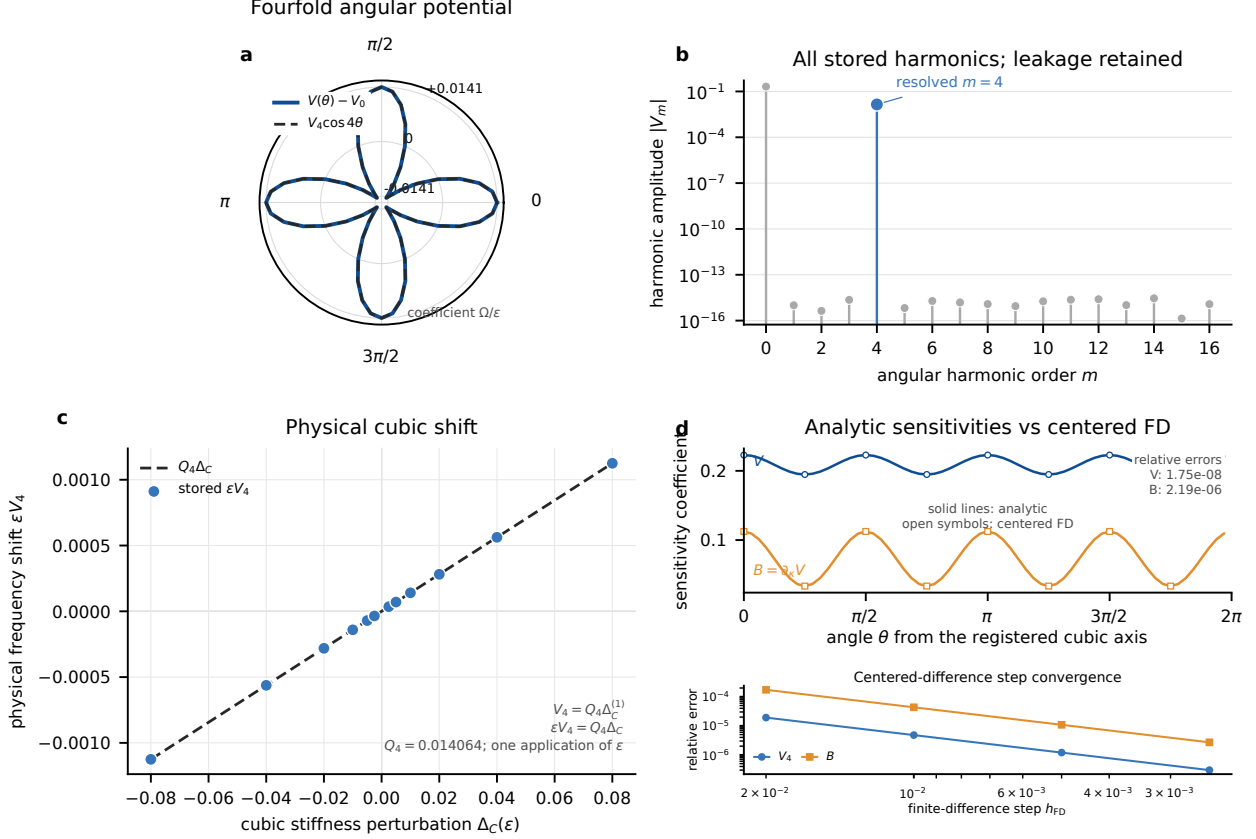


Figure 3: Full-elastic sensitivities for the declared weak cubic family. **a**, The solid curve is the analytic full-elastic coefficient $V(\theta) - V_0$, and the dashed curve is $V_4 \cos(4\theta)$. A common radial offset is used only to render the polar plot; the radial tick labels give the centered coefficient in units of Ω/ϵ . **b**, Stems and circular markers show every stored harmonic amplitude $|V_m|$ on a logarithmic scale; the $m = 4$ marker is highlighted and non-fourfold leakage is retained. **c**, Filled circles show the stored physical shift ϵV_4 , while the dashed line shows $Q_4 \Delta_C$. Their agreement checks the perturbation convention $V_4 = Q_4 \Delta_C^{(1)}$ and $\Delta_C = \epsilon \Delta_C^{(1)}$, rather than providing an independent full-wave comparison. **d**, Solid lines are the analytic full-elastic V and $B = \partial_{\kappa} V$; open circles and squares are independent centered full-wave finite differences. The displayed relative errors are $\|V_{FD} - V\|_2 / \|V\|_2$ and $\|B_{FD} - B\|_2 / \|B\|_2$. The inset gives the declared centered-difference step convergence of the relative V_4 and B errors. Source data: [data/source_data/figure.03](#).

the locations and alternating types. This is a finite-resolution local realization on one branch, not a proof of exhaustive continuum enumeration. In particular, the winding and grid checks cannot exclude an unresolved opposite-index pair between candidate nodes. Exact four-plus-four cardinality is reserved for the sufficiently-small theorem above, while the computed result is reported as eight refinement-stable roots.

The artifact field `morse_index` stores the gradient, or Poincaré, index rather than the number of negative Hessian eigenvalues: each resolved minimum contributes +1, each saddle contributes -1, and the total is zero. The outer-minus-inner gradient winding agreed with this point-index sum under Eq. (172), while both annular boundaries remained noncritical. Figure 4 displays the stored refined coordinates, Hessian inertia, and perturbative location errors. Point type was assigned only when the Cartesian Hessian signs exceeded their discrepancy estimate; it was not inferred from the appearance of interpolated contours.

3.4 Scaling and sign checks

Within the predeclared weak-anisotropy sequence, compensated full-wave data approached the radial prediction in Eq. (81), and the minimum–saddle critical-frequency separation followed Eq. (102). The stored log–log slope of the separation was 0.995817, consistent with first-order scaling, while the frequency remainder slope was 1.99957, consistent with a second-order residual. These statements are restricted to the resolved nonzero coefficients and the declared perturbation sequence; they are not extrapolations to a finite material anisotropy.

The compensated radial shifts and frequency remainders provide the primary coefficient-level checks because they retain the predicted limits directly. In particular, division by the signed perturbation approaches the full-elastic coefficient ratio $-B(\theta)/a$ on both angular families. The fitted slopes are secondary summaries of those same predeclared data, not standalone evidence or a plot-selected scaling window. Sign reversal supplied an independent control: changing εV_4 exchanged the angular roles without changing the minimum and saddle counts. Figure 5 reports the complete sequence, the first- and second-order compensated quantities, and the role-reversal check.

4 Coefficient-Resolved Temporal Crossover

4.1 Joint-limit Bessel law

Within the declared joint-limit window for the nonnodal source and angular weight class, the undoubled positive-frequency response $G^+(t; \varepsilon)$ of the tracked branch collapsed onto the coefficient-resolved Bessel form in Theorem 3.1. Equation (127) retains the radial $t^{-1/2}$ factor and angular kernel $J_0(\varepsilon V_4 t)$, with the full-elastic coefficient $V_4 = 0.0140638$. Here $t := t_{\text{phys}} c_T / h$ is dimensionless, and V_4 is the correspondingly scaled sensitivity coefficient; εV_4 is the finite- ε shift in this convention, and its dimensional value is $(c_T/h)\varepsilon V_4$. The complex collapse in Fig. 6 was evaluated without fitted alignment of amplitude, phase, frequency, or time. This connects the established isotropic radial decay and anisotropic splitting without claiming a new Bessel identity [4, 5, 7, 11, 19].

The agreement included the sign-preserving transition coordinate $\tau = \varepsilon V_4 t$, although the axisymmetric J_0 kernel is even in that coordinate. The early declared envelope fit gave the slope -0.518693 , consistent with the radial stationary-phase exponent before the fourfold angular stationary points were resolved. Scaled absolute complex error, rather than relative error, was retained around zeros of the Bessel factor. The angular integral producing J_0 is an established Bessel identity; the result here is its coefficient-resolved realization for the selected Lamb branch, not a new identity.

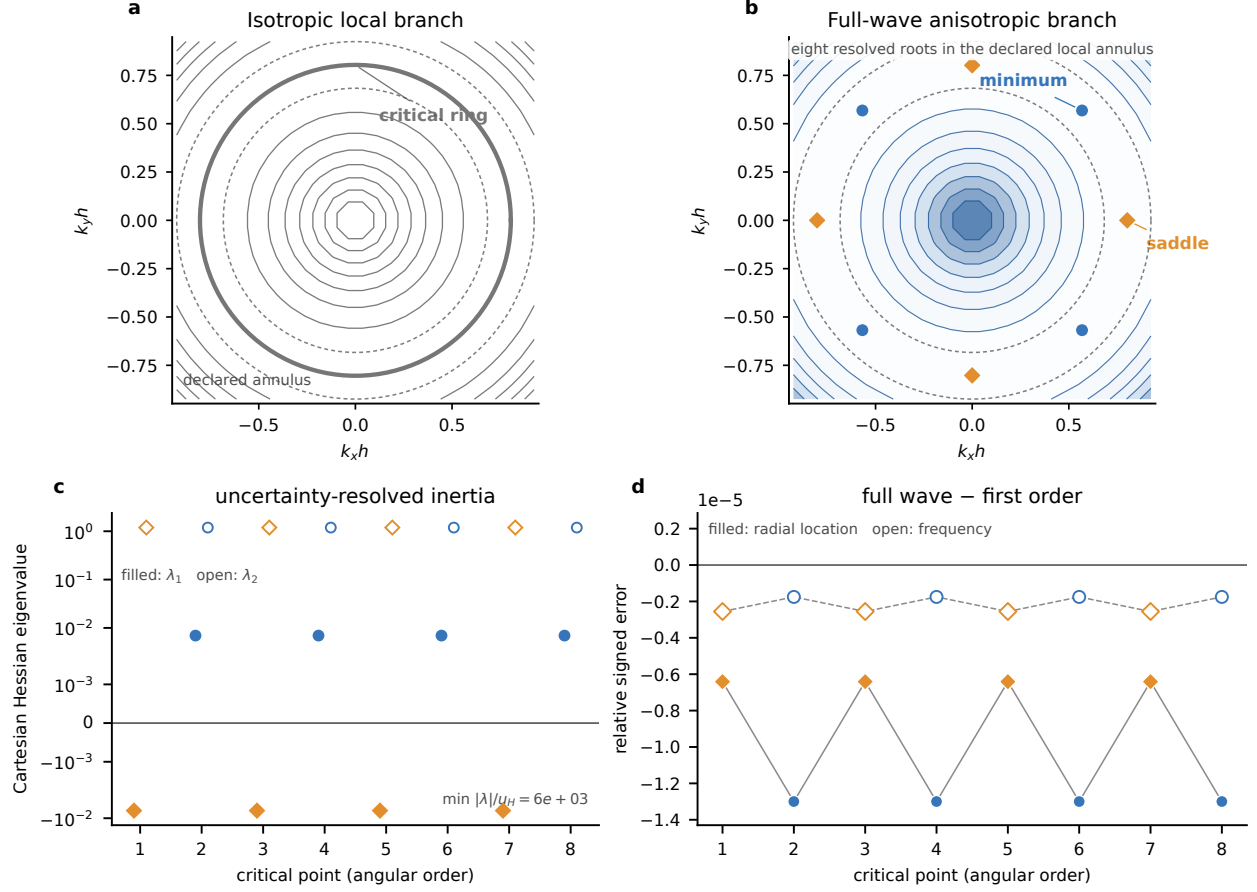


Figure 4: Full-wave numerical realization of the local Morse splitting with eight resolved roots. **a**, Grey contours show the isotropic full-wave surface, the solid blue curve marks the exact isotropic critical ring, and the two short-dashed circles delimit the declared annulus. **b**, Filled and line contours show the anisotropic full-wave surface using the same levels as panel **a**; blue circles and orange diamonds are the stored refined minima and saddles, and the short-dashed circles again show the annular boundaries. **c**, Cartesian Hessian eigenvalues are ordered by critical-point angle on a symmetric logarithmic axis. Marker shape and colour identify Morse class, filled and open symbols denote λ_1 and λ_2 , vertical bars give the declared Hessian-discrepancy estimate, and the shaded band spans the maximum such estimate about zero. **d**, Filled symbols joined by the solid guide line give the signed radial error $(\kappa_{\text{full}} - \kappa_{\text{pert}})/\kappa_0$; open symbols joined by the dashed guide line give the signed frequency error $(\Omega_{\text{full}} - \Omega_{\text{pert}})/\Omega_0$. The perturbation values are first-order predictions, and neither guide line is a fit. All surfaces and coordinates use the normalized kh and $\omega h/c_T$ convention. Source data: `data/source_data/figure_04`.

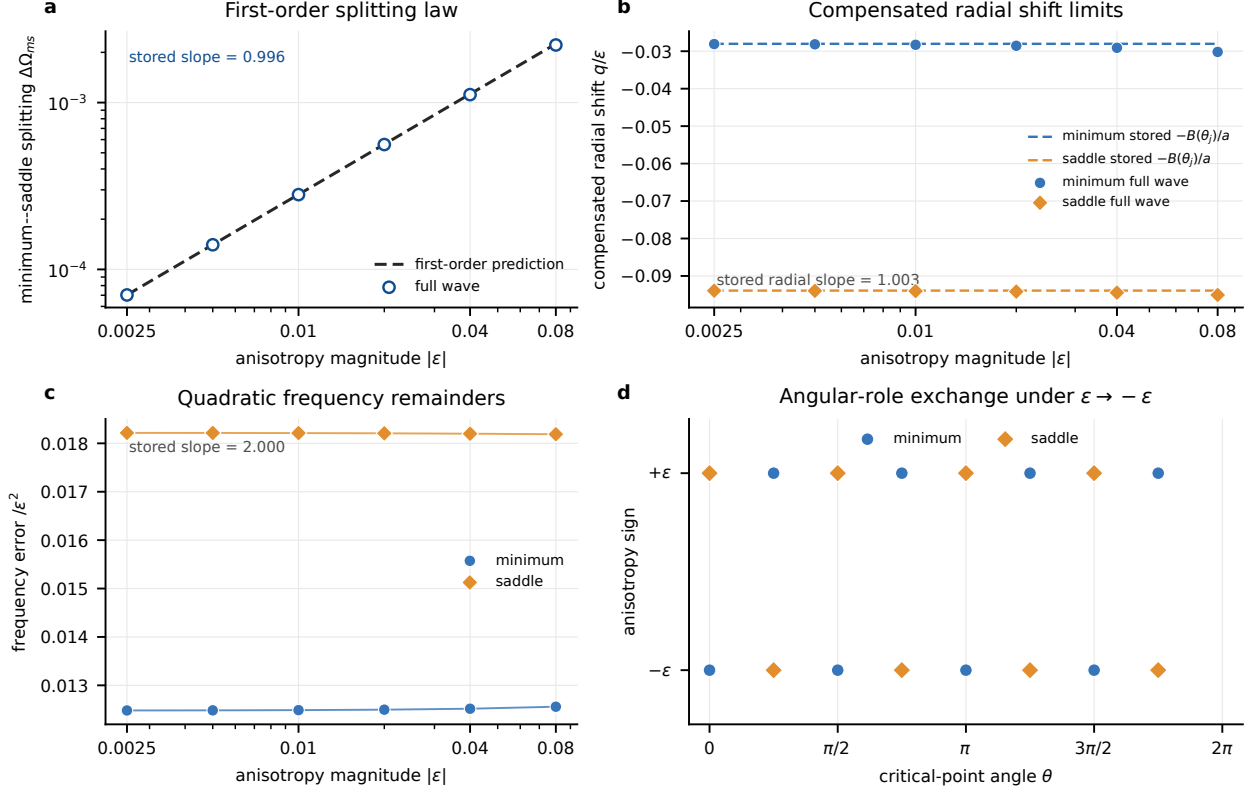


Figure 5: Perturbation checks over the complete declared weak-anisotropy sequence. **a**, Open circles are the full-wave minimum--saddle frequency splitting, and the dashed line is its first-order coefficient prediction. The reported splitting slope is an unweighted log--log fit over every declared positive $|\varepsilon|$, not a plot-selected range; no separate fitted line is drawn. **b**, Blue circles and orange diamonds show the baseline-corrected full-wave radial shifts $q_{\min}/\varepsilon$ and $q_{\text{sad}}/\varepsilon$, while the corresponding dashed lines are the coefficient limits $-B(\theta_j)/a$. The annotated radial slope is obtained from the uncompensated quantity $\max(|q_{\min}|, |q_{\text{sad}}|)$ over the same complete sequence. **c**, Markers and connecting guide lines show the absolute full-wave minus first-order critical-frequency errors divided by ε^2 for the minimum and saddle families. The reported remainder slope is fitted to the maximum uncompensated error across the two families; the connecting lines are not fits. **d**, Circle and diamond markers show the minimum and saddle angular orbits for opposite perturbation signs, demonstrating exchange of the axis and diagonal roles under $\varepsilon \rightarrow -\varepsilon$. Source data: `data/source_data/figure_05`.

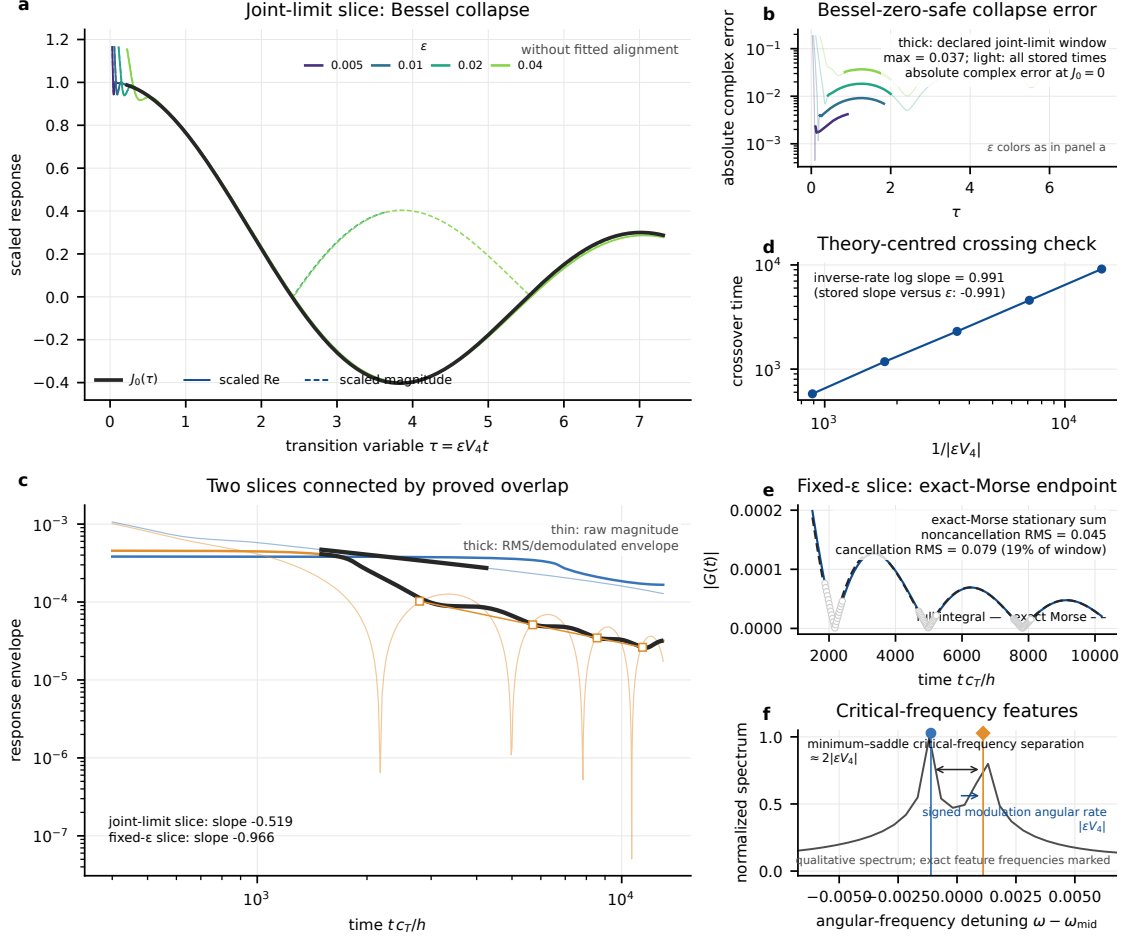


Figure 6: Two numerical asymptotic slices of the coefficient-resolved response G^+ , with $t = t_{\text{phys}} c_T/h$. **a**, Full-wave real parts and magnitudes, after removal of the shifted carrier $\Omega_0 + \epsilon V_0$ and uniform scaling, collapse onto $J_0(\tau)$, $\tau = \epsilon V_4 t$, for the four declared uniform transition rows. The comparison is performed without fitted alignment. **b**, Thin curves show the absolute complex error at all stored times; thick segments show the uniform transition window $t \geq 1500$, $|\tau| \leq 2$, including Bessel zeros. **c**, The early fit uses the raw $\epsilon = 0.005$ magnitude, whereas the late fit uses four complete beat-period RMS values in the fixed-anisotropy $\epsilon = 0.08$ row. These are two slices, not one numerical time trajectory; the proved growing- $|\tau|$ overlap connects their coefficients and phases. **d**, Crossover markers are first downward crossings of magnitude 0.9 within a theory-centred bracket. This is a theory-centred consistency diagnostic; panel **a** is the primary numerical test. **e**, At $\epsilon = 0.08$, the full integral and exact-Hessian Morse sum are compared over $1500 \leq t \leq 10200$; open markers denote cancellation samples with coherence below 0.30. **f**, Exact minimum and saddle frequencies mark the critical-frequency features; arrows distinguish their separation $2|\epsilon V_4|$ from the modulation magnitude $|\epsilon V_4|$. Source data: [data/source_data/figure_06](#).

4.2 Overlap with isolated stationary points

Under the growing- $|\tau|$ overlap conditions, but only while the accumulated second- and higher-order phase corrections remain controlled, the same branch entered the isolated-point scale $O(|\varepsilon V_4|^{-1/2}t^{-1})$. This conclusion follows from the independent growing-parameter stationary-phase argument in Theorem 3.2; the compact- τ theorem cannot be extrapolated to large $|\tau|$. The limit and its explicit error budget are stated in Eqs. (141) and (143), while Eq. (142) restores the response normalization. The overlap proof directly resolves the alternating angular stationary points and recovers the Cartesian minimum and saddle signature factors within the declared local annulus.

Expanding the large-argument Bessel factor in this controlled overlap produced the same coefficient and phases as the corresponding first-order four-minimum/four-saddle Morse sum in Eq. (150), not merely the same decay exponent. The fourfold multiplicity of each critical family cancelled the angular-curvature factor, leaving the expected radial and angular stationary-phase prefactor. Away from destructive-cancellation times, a fixed small nonzero ε slice has the t^{-1} time exponent. Along a joint path, however, the prefactor $|\varepsilon V_4|^{-1/2}$ remains explicit and can diverge as the ring is restored. Root-mean-square binning is therefore the suitable cancellation-safe observable. Figure 6 presents two numerical asymptotic slices rather than one time trajectory: a small-perturbation joint-limit slice and a fixed-anisotropy slice. The plotted late bins and -0.965731 in Fig. 6, however, belong to the separate fixed-anisotropy endpoint discussed next; they are not a numerical joint-path test of the growing- $|\tau|$ theorem. The connection between the slices is the proved growing- $|\tau|$ overlap above.

4.3 Fixed-anisotropy Morse endpoint

At fixed nonzero anisotropy, and only for resolved nondegenerate critical points away from the window boundary, standard two-dimensional stationary phase applied to the exact full-wave Cartesian branch gave the endpoint in Theorem 3.3. The signed Hessian sum in Eq. (152) uses the exact critical frequencies, modal amplitudes, and Cartesian determinants rather than their first-order approximations. It predicts a generic t^{-1} contribution to the undoubled positive-frequency response from each nonnodal isolated point; the physical real response is $2 \operatorname{Re} G^+$. This use of standard two-dimensional stationary phase follows established guided-wave asymptotics [6, 9, 21].

At fixed anisotropy, the full-wave response approached this parameter-determined sum over the declared comparison window. Its root-mean-square envelope gave the slope -0.965731 , as shown in Fig. 6. This endpoint complements the joint-limit Bessel law: the former keeps ε fixed while time increases, whereas the latter takes a controlled two-parameter limit. The remainder $O_\varepsilon(t^{-2})$ in Eq. (152) is not uniform as $\varepsilon \rightarrow 0$, so this endpoint does not replace the coalescing-ring description.

4.4 Frequency separation and modulation rate

Within the fourfold family on the tracked branch, the exact minimum and saddle frequencies appeared as two critical-frequency features of the infinite, lossless continuum. Their nonnegative separation in Eq. (155) is twice the modulation angular-frequency magnitude in Eq. (156). This factor-of-two statement is restricted to the signed two-family modulation and keeps the feature separation distinct from the magnitude that multiplies the dimensionless time t after the mean carrier has been removed.

At leading order, the separation is $2|\varepsilon V_4|$, whereas the modulation magnitude is $|\varepsilon V_4|$. The minimum and saddle signature phases then generate the signed cosine recovered in the growing-parameter overlap. The quantitative comparison uses the exact critical-point frequencies; the

spectrum panel of Fig. 6 is qualitative. At fixed anisotropy, unequal exact amplitudes and higher-order phase corrections may deform the cosine into a two-phaser modulation, but they do not alter the definition of the exact critical-frequency separation.

5 Full-Wave Numerical Verification

5.1 Convergence and resolution checks

Within the declared local annulus and on the tracked branch, the isotropic coordinates and curvature passed the predeclared order-convergence gates. The eigenpairs and continuation history separately passed the residual, orthogonality, overlap, and eigengap gates. The independently computed values remained consistent with $\kappa_0 = 0.804217$, $\Omega_0 = 2.85176$, and $a = 1.19686$, while Fig. 2 exposes the full order sequence rather than only its finest member. This evidence concerns the selected local branch and does not assert completeness for the dispersion spectrum outside the configured window.

No coordinate match was accepted alone. A converged row also had to satisfy the normalized residual of Eq. (164), the mass orthogonality defect, the mass-weighted overlap of Eq. (166), and the relative separation gate in Eq. (168). The generated Supplementary convergence table reports the final one- and two-element values beside their declared gates. Algorithm 167 supplied the branch history, so apparent coordinate agreement after a branch switch could not pass this joint check [1, 12].

The anisotropic evidence closed a distinct joint gate. Centered full-anisotropic differences checked $V_4 = 0.0140638$ and the differentiated-mode equation, while Hessian discrepancy estimates and candidate-grid refinement checked the 4-minimum and 4-saddle set in Figs. 3 and 4. The two resolution routes are defined in Eqs. (169) and (171). Boundary noncriticality and the index identity of Eq. (172) then closed the declared local annulus. Together these checks support a resolved local set on one branch, not a global enumeration or laboratory evidence.

5.2 Response and phase accuracy

For the declared joint-limit rows and the separate fixed-anisotropy row, independent full-wave polar-grid refinements converged while the maximum accumulated inter-resolution phase-discrepancy estimator remained 1.28586×10^{-6} , below its declared acceptance threshold. Each production grid rebuilt and retracked its own spectral surface before applying the Cartesian-measure quadrature in Eq. (173). The nested subsampling stored in the auxiliary convergence artifact is therefore a diagnostic only, not a substitute for these independent production surfaces.

The accumulated inter-resolution phase-discrepancy estimator in Eq. (175) combines the local nested-order frequency difference with the common-node difference between successive polar grids. It restricts the accepted reported window by comparing computed resolutions; without a continuum enclosure, it does not bound the unknown continuum phase error. Figure 6 shows that the complex response and fixed-anisotropy comparison passed this same declared consistency gate.

The masks fixed before response evaluation in Eq. (174) gave an early decay slope -0.518693 and a late decay slope -0.965731 . The early fit uses only the smallest perturbation row before angular stationary points are resolved, whereas the late fit uses complete half-open root-mean-square modulation bins in the fixed-anisotropy row. The fitted log-log intercept is a nuisance parameter for exponent estimation; it is not an adjustment of the transition or exact-Morse response curves, which use no fitted alignment.

5.3 Source and window robustness

The declared robustness sweep varied the Gaussian source radius only within its declared range. A separate sweep varied the width of the local window in wavenumber. The response norm changed by a bounded amount, while the baseline normalization and localization at the widest window boundary were preserved. Every declared pair was evaluated from the stored branch surface. This check concerns source and truncation sensitivity, rather than a new decay exponent or critical-point completeness. The full sweep and its boundary-weight diagnostic are reported in the Supplementary Information.

The baseline row is included in both one-parameter sweeps, so every reported difference has a common normalization. The window check also bounds the weighted contribution at the radial endpoints, limiting the possibility that the response trend is created by abrupt branch truncation. These controls support the interpretation of Fig. 6 without replacing the asymptotic hypotheses or the independent phase-discrepancy check.

5.4 Silicon stress test

A separate calculation for a $[001]$ -cut silicon plate recovered an alternating local minimum–saddle set with resolved Cartesian Hessians and noncritical annular boundaries, serving only as a finite-anisotropy stress test. The calculation repeats the tracked-branch search and the winding procedure of Eq. (172), but physical silicon lies outside the controlled weak-anisotropy sequence used for the coefficient and remainder claims [11, 20].

Accordingly, this result is reported as numerical robustness of the search, classification, and boundary-check machinery under a materially larger constitutive contrast. It supplies neither evidence for the weak-parameter scaling laws nor a count over the complete silicon dispersion surface. The local alternating organization is consistent with earlier anisotropic ZGV computations, while the claims of this study remain attached to the declared perturbative family and its explicitly tracked branch. The corresponding contours, refined points, Hessian inertia, and boundary margin are reported in the Supplementary Information.

6 Discussion

6.1 Geometry and noncommuting limits

Within the controlled weak-cubic family and the declared local annulus, the essential change is geometric rather than a uniform frequency displacement. The local Morse–Bott ring unfolds into isolated Morse minima and saddles. The isotropic ZGV critical ring is Morse–Bott: radial localization is nondegenerate, but continuous rotation leaves a tangential Hessian zero mode (Theorem 1.1). Cubic anisotropy lifts that zero mode and organizes the local stationary set into alternating Cartesian Morse points (Theorem 2.2). The extra nondegenerate direction adds angular stationary phase to the existing radial stationary phase. This explains why the ring and point regimes carry different time exponents without describing the change as a phase transition.

The two exponents belong to noncommuting limits. Write $\alpha = \varepsilon V_4$. The ring-scale response is proportional to $t^{-1/2} J_0(\alpha t)$, whereas the isolated-point scale in the controlled overlap is $|\alpha|^{-1/2} t^{-1}$. The latter prefactor follows from the tangential Hessian curvature in Eq. (148); point multiplicity cancels the fourfold curvature factor but not $|\alpha|^{-1/2}$. Its growth as $\alpha \rightarrow 0$ signals loss of uniformity when the points coalesce back into a ring, not divergence of the physical solution. At $t \sim |\alpha|^{-1}$, both asymptotic scales are of order $|\alpha|^{1/2}$, as summarized in Eq. (158).

The joint-limit Bessel law and fixed-anisotropy exact-Morse sum are therefore complementary. The uniform Bessel theorem controls compact $\tau = \alpha t$ (Theorem 3.1); its large-parameter edge is matched only through the separately proved growing- $|\tau|$ overlap and Eq. (142). At fixed nonzero anisotropy, the exact-Morse theorem instead uses full critical frequencies, amplitudes, and Cartesian Hessians (Theorem 3.3). Its remainder is not uniform as $\varepsilon \rightarrow 0$, while a first-order Bessel phase is not indefinitely accurate at fixed ε . Neither formula can be substituted for the other outside the regime it controls.

6.2 Relation to prior work

Within the scope of this comparison, both endpoint geometries are established results rather than discoveries of the present study. The isotropic ZGV critical ring and its $t^{-1/2}$ response follow the classical Lamb-wave setting [14, 18, 19]. Directional anisotropic shifts were also established [20], and isolated anisotropic ZGV points were previously reported. In particular, the four minima and four saddles were previously reported together with their interference, and beating responses were already reported for cubic silicon [11, 13]. Full-spectrum algorithms for locating such points are likewise prior infrastructure [12].

The distinction of the present calculation is the controlled, coefficient-resolved link between those known endpoints. Morse–Bott classification and perturbation are established mathematics [2, 3]. Here transverse nondegeneracy is verified on the exact Lamb determinant and independently on the tracked elastic branch. The fourfold coefficient is obtained from the full generalized-eigenvalue sensitivity, and the radial coefficient retains the differentiated-mode contribution. Finite differences, compensated radial shifts, remainder scaling, sign reversal, Cartesian Hessian inertia, and annular index closure then test the reduction. Thus the local four-plus-four result follows from computed coefficients and controlled remainders, not from point-group symmetry alone or from fitting the final critical locations.

The asymptotic tools also have clear precedents. Broken-symmetry uniform approximations with Bessel modulation connect continuous families and isolated objects in other wave settings [4, 5, 7]. Two-dimensional stationary phase is standard and has been used for anisotropic guided-wave radiation [6, 9, 21]. Accordingly, the contribution is neither a new Bessel identity nor a new stationary-phase theorem. It is their fully normalized Lamb-wave realization with an elasticity-derived Bessel argument, an explicit phase-validity window, and a constant-and-phase match to the first-order Morse overlap. Separately, the fixed-anisotropy endpoint is compared numerically with the exact-Hessian point sum without fitted amplitude, phase, frequency, or time shifts.

6.3 Scope and boundary statements

The canonical Bessel law holds only within the specified nonnodal source and angular-weight class. A general source and its angular weight jointly produce the Fourier–Bessel series of Eq. (134); the leading kernel need not reduce to a single J_0 term. If the ring amplitude vanishes, or if a detector is nodal at a particular Morse point, the leading contribution changes as described in Supplementary Section 3.8. Cancellation times do not alter the generic exponent, but they do make a pointwise relative error or local slope ill conditioned. The declared source-radius and window sweeps support robustness only over their declared ranges, not source independence.

The generality of the result has two levels. At theorem level, any simple, uniformly gapped finite-radius branch with nonzero radial curvature and a dominant nonzero fourfold perturbation coefficient has the same local Morse–Bott-to-Morse structure and exact sufficiently-small four-plus-four count under the stated remainder hypotheses. A pure leading fourfold phase, together with the

declared nonnodal source and phase-remainder conditions, is additionally required for the single- J_0 law and its matched asymptotic regimes; generic higher harmonics change the angular canonical integral. At instance level, coefficient values, finite-resolution roots, and response comparisons are demonstrated for one selected Lamb branch and one cubic perturbation family. The silicon stress test assesses only the search and classification workflow.

The geometric count is also restricted to one simple branch in the declared local annulus. Branch contamination is controlled here by tracking and the eigengap gate of Eq. (168); if the eigengap closes, a scalar sensitivity and reduced resolvent no longer suffice. Degenerate, matrix-valued perturbation theory would then be required. Likewise, zero radial curvature, a zero-wavenumber stationary point, or a singular exact Hessian would demand a different canonical scaling. The computed local count does not enumerate every Lamb or shear-horizontal branch.

Fourfold symmetry alone does not fix the number of points. The pure leading fourfold term is a result of the declared first-order cubic family. If the coefficient V_4 is zero, higher harmonics or the second-order effective angular potential can control the splitting; competing higher harmonics can also change the root count. The coefficient B is not part of the controlled remainder: the explicit $\varepsilon q B$ term controls the radial shift, and B enters the phase-error and overlap budget through the effective second-order phase. Omitting B therefore changes both the critical locations and temporal validity. These cases require a new coefficient audit rather than extrapolation of the present four-plus-four formula.

Neither temporal approximation is indefinitely phase accurate outside its declared late-time window. Compact- τ collapse, the growing-parameter overlap, and the fixed-anisotropy comparison impose different error conditions. The accumulated inter-resolution phase-discrepancy estimator in Eq. (175) restricts the accepted numerical time window but does not bound the unknown continuum phase error. The operational crossover time in Fig. 6 is found inside a theory-centred bracket and is therefore a consistency diagnostic, not independent evidence for inverse-rate scaling; the complex Bessel collapse is the primary numerical test. Critical-frequency features in the lossless continuum should not be interpreted as discrete resonances, and their exact separation remains twice the defined modulation magnitude.

Finally, the present model is a linear, infinite, homogeneous, lossless plate. Damping generally introduces complex dispersion and additional decay; finite boundaries introduce reflections and a discrete modal spectrum; layers, interfaces, and defects add branches, scattering, and new angular weights. Each extension requires renewed branch projection, coefficients, and phase-error analysis. The finite-anisotropy silicon stress test is only a robustness check on search, classification, and boundary consistency checking. It supplies no weak-parameter theorem and no global point count. This computation-only study includes no experiments.

6.4 Nonlinear amplitude equations are a separate problem

The normal form used here is only a linear spectral reduction, not a nonlinear amplitude equation. Its frequency sensitivities and dispersive exponents describe propagation after an impulsive load under the linear constitutive model. They do not contain nonlinear coupling, saturation, drive-dependent detuning, mode competition, or pattern selection. In particular, the local Morse geometry can organize the linear spectral input to a later theory, but it does not determine nonlinear states or their stability.

Nonlinear amplitude equations are therefore a separate problem. Such a study would require a nonlinear constitutive law, nonlinear traction conditions, a multiscale solvability calculation, and declared scalings for forcing, damping, and detuning. Resonant projections and coupling coefficients would have to be derived independently, followed by stability analysis. None of those

equations or coefficients is obtained from the present Green response, and no nonlinear saturation or pattern-selection result is inferred here.

7 Methods

7.1 Governing equations and nondimensionalization

We considered an infinite homogeneous elastic plate occupying $\mathbb{R}^2 \times [-h, h]$, with density ρ , elasticity tensor C_{ijkl} , and traction-free faces. For a plane-wave component $\mathbf{u} = \mathbf{U}(z) \exp[i(k_x x + k_y y) - i\omega t]$, the strong problem is $\partial_j(C_{ijkl}\partial_l u_k) = \rho \partial_t^2 u_i$, with $\boldsymbol{\sigma}\mathbf{n} = 0$ at $z = \pm h$. The reference isotropic solid has $\lambda = 2$, $\mu = 1$, $\rho = 1$, and $h = 1$. The shear speed is $c_T = \sqrt{\mu/\rho}$; reported wavenumber, frequency, and time are $\kappa = hk^{\text{phys}}$, $\Omega = h\omega^{\text{phys}}/c_T$, and $t = c_T t^{\text{phys}}/h$. Consequently, every plotted coordinate and fitted time is dimensionless. The signed parameter ε multiplies the declared cubic constitutive perturbation at fixed density; it is not a frequency shift.

The isotropic symmetric branch was first computed from the desingularized Rayleigh–Lamb determinant, rather than from a plate approximation. Its first S_1 -ZGV root in the declared search window was obtained at 80-decimal working precision from the simultaneous determinant and radial-derivative equations. Implicit differentiation supplied the radial curvature. The selected root and curvature then served as independent regression targets for the thickness-resolved solver. The full determinant construction, desingularization at the shear cutoff, and Morse–Bott Hessian calculation are given in Supplementary Section 1.

7.2 Spectral solvers

The three displacement components were discretized only through the thickness by a Legendre–Gauss–Lobatto (GLL) spectral element method. The weak form retains the face traction as its natural boundary term; no face degree of freedom is constrained and no penalty is added. Stiffness and mass were assembled by collocated GLL quadrature in the Mandel strain convention. The production homogeneous-plate mesh uses one element on $[-h, h]$; a two-element mesh split at $z = 0$ is an independent assembly check. The generalized Hermitian problem $\mathbf{K}\mathbf{u} = \omega^2 \mathbf{M}\mathbf{u}$ was solved for the lowest 12 modes with SciPy’s generalized-Hermitian `gvx` driver. Eigenvectors were mass normalized. Matrix Hermiticity, mass orthogonality, and the normalized eigen residual were tested at every declared convergence order.

All first and second derivatives of $\mathbf{K}$ with respect to (k_x, k_y) were assembled analytically from the wavevector-affine strain matrix. Absolute frequency and gradient discrepancy estimators were the differences between answers at orders p and $p + 4$. These inter-order quantities are empirical resolution indicators, not bounds on the unknown continuum errors. The relative estimator used by the eigengap gate was the maximum of that nested-order eigenvalue discrepancy and the two eigen residuals. The committed full profile uses order pairs 12/16 for the isotropic artifact, 10/14 for the critical-point, scaling, and Green artifacts, and 12/16 for the silicon stress test; convergence sweeps the pairs 6/10 through 14/18. The sensitivity artifact uses order 10 and is checked by independent physical finite differences rather than a 10/14 pair. The exact weak form, derivative matrices, residual definitions, and all fixed gates are specified in Supplementary Section 4.

7.3 Mode tracking and sensitivities

Parity identifies the symmetric branch in the independent isotropic validation. Production angular anchors were seeded by proximity to the exact target frequency and then continued by a mass-

weighted modal assurance criterion. Near a close pair, tracking used the complete connected eigenvalue cluster and its principal subspace overlap before phase alignment. Eight endpoint-free angular anchors were constructed independently at orders p and $p + 4$; closure of the angular loop required both scalar MAC and the least principal cosine to be at least $1 - 10^{-8}$. Inward and outward radial rays were continued separately from each ring anchor. Samples were rejected when the selected cluster touched the top of the 12-mode solve or when the relative eigengap was no more than ten times the relative eigenvalue discrepancy estimator. This eigengap gate is a method hypothesis: no scalar sensitivity or Hessian was accepted through a numerically unresolved crossing.

The generalized-eigenvalue sensitivity was evaluated at fixed mass norm by the Hellmann–Feynman contraction. The angular function $V(\theta)$ and its mean and fourth harmonic, V_0 and V_4 , were computed on 64 endpoint-free angles. The mixed radial coefficient B retained the differentiated-mode contribution. The mode derivative was obtained from a bordered reduced-resolvent system with the mass-gauge constraint, and both the equation and gauge residuals were required to be below 10^{-8} . A centered full-wave difference in ε checked $V(\theta)$, and a mixed (ε, q) difference checked $B(\theta)$, both with $\Delta\varepsilon = \Delta q = 0.0025$. An independent tensor-product step sweep 0.02, 0.01, 0.005, 0.0025 checked convergence. Supplementary Section 2 gives the complete generalized-pencil algebra, rotated cubic contraction, normal form, and Morse splitting proof.

7.4 Critical-point search and annular consistency check

Critical points were searched only on the tracked branch inside the annulus $(1 - f_A)k_0 \leq |\mathbf{k}| \leq (1 + f_A)k_0$, with $f_A = 0.15$. Every polar cell in which both radial and tangential gradient components bracketed zero supplied a seed. Each seed was refined in Cartesian coordinates by `scipy.optimize.root` with the default Powell hybrid method, a Richardson Cartesian Hessian from steps h_H and $h_H/2$ as Jacobian, and solver tolerance 10^{-11} . A root outside the annulus or with a gradient residual exceeding its nested-order gradient-discrepancy estimator was rejected. Roots were merged only within the propagated, capped position-resolution radius.

The full calculation repeated the candidate search on 5×16 and 9×32 radial-by-angular grids. The Hessian was Richardson extrapolated from centered gradient differences with full-profile step 10^{-3} ; a classification was accepted only if its least absolute eigenvalue exceeded ten times the Hessian discrepancy estimate. Counts, kinds, locations, frequencies, and Hessian eigenvalues had to agree under candidate-grid doubling. An independent annular check sampled each boundary at 32 and 64 angles, required a gradient norm separated from its discrepancy estimate, stable boundary winding, angular increments below $\pi/2$, and equality between outer-minus-inner winding and the sum of point indices. The $\varepsilon < 0$ search was repeated on an angularly offset candidate grid. This is a finite-resolution local-annulus consistency check, not an exhaustive continuum enumeration and not a global enumeration of every Lamb and shear-horizontal branch. Index closure alone also cannot exclude an unresolved pair of opposite-index critical points between candidate-grid nodes; candidate-grid doubling, symmetry-offset repetition, and Hessian separation supply the additional controls used here.

A separate finite-anisotropy stress test used crystal axes $x \parallel [100]$, $y \parallel [010]$, and $z \parallel [001]$, so the plate normal was $[001]$. The constants $C_{11} = 165.7$, $C_{12} = 63.9$, and $C_{44} = 79.6$ GPa followed the published single-crystal elasticity record [22]. The calculation normalized stiffness by C_{44} , set $\rho = h = 1$, and reported frequency in $\sqrt{C_{44}/\rho}/h$. It exercised the search and annular consistency check under stronger constitutive contrast and was not used to validate the weak- ε theorem.

7.5 Green-function quadrature and fit masks

The positive-frequency response was projected onto the tracked branch for a normal impulsive source and top-face normal detector. The modal amplitude was $i|u_z(h)|^2/(2\omega)$. A Gaussian source factor with $R/h = 0.5$ and a fixed branch window $\exp[-(q/\sigma)^8]b_\sigma(q)$, where $q = |\mathbf{k}| - k_0$ and $\sigma/k_0 = 0.15$, were applied. The deterministic C^∞ taper b_σ equals one for $|q| \leq 1.25\sigma$, uses the standard flat $\exp(-1/x)$ transition, and is exactly zero for $|q| \geq 1.5\sigma$. Thus the integration interval $[-1.5\sigma, 1.5\sigma]$ contains the complete compact support and has no endpoint contribution. Direct integration used the Cartesian Fourier measure with radial polar Jacobian $k_0 + q$, an endpoint-free trapezoidal rule in angle, and composite Simpson quadrature in radius. No Bessel function entered this full-wave quadrature. The full profile rebuilt independent tracked surfaces on 129×32 , 257×64 , and 513×128 grids for the uniform comparisons at $\varepsilon \leq 0.04$; the fixed $\varepsilon = 0.08$ comparison used twice as many angular nodes. For every response row, the stationary-point search used an annular half-width $1.55\sigma = 0.2325k_0$, strictly larger than the direct support $1.5\sigma = 0.225k_0$, on a 15×32 candidate grid. Boundary/index checks used 16 and 32 angular nodes. Every resolved stationary point in this larger domain was evaluated with the same compact taper and included in the fixed-Morse sum; the domain, margin, point counts, and contribution counts are stored in the Green sidecar. The computed time axis was $t_n = 400 + 25n$, $0 \leq n \leq 504$, for $\varepsilon \in \{0.005, 0.01, 0.02, 0.04, 0.08\}$.

Fit masks were fixed in code before response evaluation. The early slope used $\varepsilon = 0.005$, $t \geq 1500$, and $0.10 \leq |\varepsilon V_4 t| \leq 0.30$. The fixed-anisotropy late slope used complete half-open minimum-saddle modulation periods beginning at $t = 1500$, with at least three samples per period and four periods. Ordinary least squares was applied to $\log t$ and the logarithm of the predeclared RMS or demodulated amplitude. The fitted intercepts were diagnostic only: the Bessel and exact-Morse comparisons used no fitted alignment of amplitude, phase, carrier frequency, or time. The full response was accepted only if the three-grid complex error contracted and the accumulated inter-resolution phase-discrepancy estimator satisfied the declared $t_{\max} \max |\delta\omega| \leq 0.05$ rad acceptance test. This estimator compares computed resolutions and does not bound the unknown continuum phase error. Supplementary Section 3 derives the normalized response and all asymptotic regimes; Supplementary Section 4 specifies the quadrature, fit masks, cancellation exclusion, and error gates.

7.6 Reproducibility

Reference physical parameters and cross-stage acceptance limits are versioned in `config/reference.yaml`; stage-specific grids, masks, and thresholds are versioned in the workflow source and repeated in artifact sidecars. The interpreter is pinned by `.python-version`, and dependency resolution is fixed by `uv.lock`. The seven workflow stages write paired NPZ arrays and JSON sidecars containing array schemas, units, commands, input hashes, source and lock hashes, tolerances, measured diagnostics, and SHA-256 output hashes. Figures read only qualified artifacts, export their plotted source CSV files, and pass a strict geometry, provenance, and typography audit.

The committed environment used `uv 0.11.19`, `Python 3.12.13`, `NumPy 2.5.1`, `SciPy 1.18.0`, `SymPy 1.14.0`, `Matplotlib 3.11.0`, `mpmath 1.3.0`, and `PyYAML 6.0.3`. Supported Python versions and dependency ranges remain declared in `pyproject.toml`; the interpreter pin and lock file, not those ranges, define the verified Python environment. PDF compilation used `latexmk 4.88`, `pdfTeX 1.40.29`, and `BibTeX 0.99e` from TeX Live 2026; these are external build prerequisites rather than Python dependencies.

From the package root, the currently executable scientific rebuild is

```
uv sync --python 3.12.13 --frozen --all-extras
uv run python scripts/reproduce_all.py \
```

```
--profile full --skip-paper
uv run python scripts/export_manuscript_values.py
```

The full-profile command first runs the complete test suite, then the isotropic, sensitivity, critical-point, scaling, Green, convergence, and silicon stages in isolated processes. It validates each artifact immediately, validates the provenance manifest, rebuilds the six main and six supplementary figures and two generated tables, and performs strict figure QA. The next command exports the evidence-checked manuscript macros. The locked compiler then checks the recorded latexmk, pdfTeX, and BibTeX versions, builds the Supplement from `supplement.tex` before the main text from `main.tex` in two independently cleaned cycles, applies the strict final-log gate, rejects nondeterministic PDF metadata, and requires the two cycles to agree byte for byte:

```
uv run python scripts/compile_paper.py
```

The supplement must be compiled first because the main document imports its label file. The smoke profile is a runtime check only; it is not the source of committed numerical evidence. Availability boundaries are stated below.

8 Conclusion

We connect a known isotropic ZGV critical ring to its known anisotropic isolated-point endpoint through a coefficient-resolved geometric and temporal reduction. Full-elastic sensitivities determine V_4 and B , while the exact determinant, independent spectral discretization, and centered finite-difference checks close the coefficient evidence. The controlled normal form gives the exact critical-point cardinality for sufficiently small perturbations, while the finite-resolution calculation resolves the corresponding alternating minima and saddles with the predicted splitting, radial shift, remainder, and sign reversal. Response quadrature that passes the inter-resolution phase-discrepancy test then connects the Bessel transition to the exact-Hessian fixed-anisotropy Morse endpoint. This connection explains how the change in stationary-set dimension and Hessian rank adds angular stationary phase and changes the computed temporal power law. The response comparison is performed without fitted alignment of amplitude, phase, frequency, or time.

The two temporal descriptions express noncommuting limits. In the joint limit $\varepsilon \rightarrow 0$, $t \rightarrow \infty$, with $\tau = \varepsilon V_4 t = O(1)$, the radially localized ring retains its $t^{-1/2}$ factor and acquires the $J_0(\tau)$ modulation. At fixed, resolved nonzero anisotropy, the isolated Cartesian Morse points instead give the ultimate t^{-1} signed-Hessian sum, whose remainder is not uniform as the ring is restored. The exact minimum-saddle critical-frequency separation is twice the modulation angular-frequency magnitude; keeping those quantities distinct prevents a frequency-feature separation from being confused with the rate that multiplies time.

The validity scope is restricted to one simple tracked branch of a lossless infinite plate, a declared local annulus, a nonnodal source and weight class, and the controlled weak-anisotropy family. The results are computation-only and make no global count across all branches; the finite-anisotropy silicon stress test lies outside the asymptotic evidence. Weak loss, finite boundaries, layered or defective plates, and near-closing eigengaps require extensions of the coefficient and error analysis. Nonlinear amplitude equations are a separate problem and belong to a distinct study rather than to the linear response developed here.

Data Availability

All data supporting the conclusions are present in the accompanying source tree. Seven qualified artifact pairs—NPZ arrays plus JSON sidecars—and the auxiliary isotropic validation and convergence records reside in `data/generated`. A separate source-data directory contains the machine-readable source data in CSV format plotted in every main and supplementary figure. Generated manuscript tables and `data/provenance_manifest.json` are also included. Artifact sidecars and the provenance manifest record SHA-256 hashes for qualified inputs and outputs. At this pre-deposition stage, no external accession, repository DOI, or release-wide checksum manifest is available.

Code Availability

The accompanying source tree contains all Python source, tests, the reference configuration, the interpreter pin, the exact dependency lock, figure scripts, and LaTeX sources required to regenerate the numerical artifacts, figures, tables, and PDFs. The current full scientific route is `uv run python scripts/reproduce_all.py --profile full --skip-paper`; the evidence macros and two PDFs are then rebuilt by the explicit commands in Methods, with the supplement compiled before the main paper. No proprietary solver or experimental data are required.

References

- [1] A. T. I. Adamou and R. V. Craster. Spectral methods for modelling guided waves in elastic media. *The Journal of the Acoustical Society of America*, 116(3):1524–1535, 2004. doi: 10.1121/1.1777871.
- [2] Augustin Banyaga and David E. Hurtubise. Cascades and perturbed Morse–Bott functions. *Algebraic & Geometric Topology*, 13(1):237–275, 2013. doi: 10.2140/agt.2013.13.237.
- [3] Raoul Bott. Nondegenerate critical manifolds. *The Annals of Mathematics*, 60(2):248–261, 1954. doi: 10.2307/1969631.
- [4] M. Brack, P. Meier, and K. Tanaka. Uniform trace formulae for SU(2) and SO(3) symmetry breaking. *Journal of Physics A: Mathematical and General*, 32(2):331–353, 1999. doi: 10.1088/0305-4470/32/2/009.
- [5] Matthias Brack and Jérôme Roccia. Closed orbits and spatial density oscillations in the circular billiard. *Journal of Physics A: Mathematical and Theoretical*, 42(35):355210, 2009. doi: 10.1088/1751-8113/42/35/355210.
- [6] Bastien Chapuis, Nicolas Terrien, and Daniel Royer. Excitation and focusing of Lamb waves in a multilayered anisotropic plate. *The Journal of the Acoustical Society of America*, 127(1):198–203, 2010. doi: 10.1121/1.3263607.
- [7] Stephen C. Creagh. Trace formula for broken symmetry. *Annals of Physics*, 248(1):60–94, 1996. doi: 10.1006/aphy.1996.0051.
- [8] Takasar Hussain and Faiz Ahmad. Lamb modes with multiple zero-group velocity points in an orthotropic plate. *The Journal of the Acoustical Society of America*, 132(2):641–645, 2012. doi: 10.1121/1.4730891.

- [9] Alexander Karmazin, Evgenia Kirillova, Wolfgang Seemann, and Pavel Syromyatnikov. A study of time harmonic guided Lamb waves and their caustics in composite plates. *Ultrasonics*, 53(1): 283–293, 2013. doi: 10.1016/j.ultras.2012.06.012.
- [10] Soufien Karous, Souhail Dahmen, Mohamed Shili Bouhdima, Morched Ben Amor, and Christ Glorieux. Multiple zero group velocity Lamb modes in an anisotropic plate: propagation along different crystallographic axes. *Canadian Journal of Physics*, 97(10):1064–1074, 2019. doi: 10.1139/cjp-2018-0348.
- [11] Daniel A. Kiefer, Sylvain Mezil, and Claire Prada. Beating resonance patterns and extreme power flux skewing in anisotropic elastic plates. *Science Advances*, 9(51):eadk6846, 2023. doi: 10.1126/sciadv.adk6846.
- [12] Daniel A. Kiefer, Bor Plestenjak, Hauke Gravenkamp, and Claire Prada. Computing zero-group-velocity points in anisotropic elastic waveguides: Globally and locally convergent methods. *The Journal of the Acoustical Society of America*, 153(2):1386–1398, 2023. doi: 10.1121/10.0017252.
- [13] Daniel A. Kiefer, Sylvain Mezil, and Claire Prada. Extreme wave skewing and dispersion spectra of anisotropic elastic plates. *Physical Review Research*, 7(1):L012043, 2025. doi: 10.1103/PhysRevResearch.7.L012043.
- [14] Horace Lamb. On waves in an elastic plate. *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, 93(648):114–128, 1917. doi: 10.1098/rspa.1917.0008.
- [15] Jérôme Laurent, Daniel Royer, and Claire Prada. Temporal behavior of laser induced elastic plate resonances. *Wave Motion*, 51(6):1011–1020, 2014. doi: 10.1016/j.wavemoti.2014.04.001.
- [16] Bor Plestenjak, Daniel A. Kiefer, and Hauke Gravenkamp. A Sylvester equation approach for the computation of zero-group-velocity points in waveguides. *Computational Mechanics*, 76(5): 1443–1456, 2025. doi: 10.1007/s00466-025-02656-8.
- [17] Claire Prada, O. Balogun, and T. W. Murray. Laser-based ultrasonic generation and detection of zero-group velocity lamb waves in thin plates. *Applied Physics Letters*, 87(19):194109, 2005. doi: 10.1063/1.2128063.
- [18] Claire Prada, Dominique Clorennec, and Daniel Royer. Local vibration of an elastic plate and zero-group velocity Lamb modes. *The Journal of the Acoustical Society of America*, 124(1): 203–212, 2008. doi: 10.1121/1.2918543.
- [19] Claire Prada, Dominique Clorennec, and Daniel Royer. Power law decay of zero group velocity Lamb modes. *Wave Motion*, 45(6):723–728, 2008. doi: 10.1016/j.wavemoti.2007.11.005.
- [20] Claire Prada, Dominique Clorennec, Todd W. Murray, and Daniel Royer. Influence of the anisotropy on zero-group velocity Lamb modes. *The Journal of the Acoustical Society of America*, 126(2):620–625, 2009. doi: 10.1121/1.3167277.
- [21] Alexander Velichko and Paul D. Wilcox. Modeling the excitation of guided waves in generally anisotropic multilayered media. *The Journal of the Acoustical Society of America*, 121(1):60–69, 2007. doi: 10.1121/1.2390674.
- [22] Lin Zhang, Raymond Barrett, Peter Cloetens, Carsten Detlefs, and Manuel Sanchez del Rio. Anisotropic elasticity of silicon and its application to the modelling of X-ray optics. *Journal of Synchrotron Radiation*, 21(3):507–517, 2014. doi: 10.1107/S1600577514004962.